HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python** backend services and **REST APIs** using **FastAPI** to support AI model inference, processing an average of **1,500 requests per hour**.
- Built and maintained **data pipelines** in **Python** for ingesting and transforming diverse datasets, including structured and unstructured text, for AI model training and evaluation.
- Integrated **language models** (e.g., via Hugging Face APIs) into applications, focusing on structured outputs, prompt engineering, and efficient handling of **context windows**.
- Designed and implemented **database schemas** and wrote complex **SQL** queries in **PostgreSQL** for data retrieval and manipulation, supporting application backends and analytical needs.
- Containerized applications using **Docker** and managed deployments on **AWS EC2** instances, ensuring consistent environments and facilitating CI/CD workflows.
- Collaborated with cross-functional teams to debug complex issues across the full development lifecycle, from prototype to deployment, improving system performance by **15%** through targeted optimizations.
- Implemented **Git** branching strategies and conducted code reviews for a team of 3 engineers, ensuring code quality and adherence to project standards.

SKILLS

Programming Languages – Python, SQL, Java

AI/ML Frameworks – PyTorch, scikit-learn, Hugging Face

Backend Development – FastAPI, Flask, Spring Boot, REST APIs

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins

Data Handling – pandas, NumPy, Data Pipelines, ETL, Data Modelling

AI Concepts – Language Models, Embeddings, Prompting, Inference, RAG, Vector Databases

PROJECTS

AI-Powered Document Analysis System

- Developed a **Python** application using **FastAPI** to process and analyze unstructured documents, integrating a **language model** for summarization and entity extraction.
- Implemented **RAG** patterns to enable the system to query and retrieve information from a custom knowledge base stored in a **vector database**, improving response accuracy by **22%**.
- Containerized the application with **Docker**, enabling seamless deployment and scaling on **AWS** infrastructure.

Customer Interaction Data Pipeline

- Engineered an end-to-end **data pipeline** using **Python** and **Pandas** to ingest, clean, and transform customer interaction logs from multiple sources.
- Developed **SQL** scripts for data validation and aggregation within a **PostgreSQL** database, preparing data for downstream analytical dashboards and AI model feature generation.
- Automated report generation for business stakeholders, delivering key insights on customer engagement patterns.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)